Stage 3C Physics Note: Symbolic Cartoon Geometry Checks

GL-with-AI Project

June 15, 2026

1 Scope and Status

This note summarizes the Stage 3C symbolic derivation scaffold for the modified-cartoon black-string reduction. The scaffold is documentation and derivation tooling only. It is not GRChombo evolution code and does not implement CCZ4 source terms.

2 Spatial Metric and Christoffels

The check starts from the four-spatial-dimensional GP spatial slice in coordinates (x, θ, ϕ, z),

$$dl^2 = dx^2 + x^2 d\theta^2 + x^2 \sin^2\theta \, d\phi^2 + dz^2. \tag{1}$$

Computing Christoffel symbols from this metric gives the angular components needed for the reduced cartoon bookkeeping:

$$\Gamma^x_{\theta\theta} = -x, \qquad \Gamma^x_{\phi\phi} = -x\sin^2\theta, \tag{2}$$

$$\Gamma^\theta_{x\theta} = \frac{1}{x}, \qquad \Gamma^\phi_{x\phi} = \frac{1}{x}. \tag{3}$$

3 GP Extrinsic Curvature

For the natural GP radial shift

$$\beta^x = s\sqrt{\frac{r_0}{x}}, \tag{4}$$

with $\beta_x = \beta^x$, $\alpha = 1$, and time-independent spatial metric, the convention checked by the script is

$$K_{ij} = \frac{1}{2}\left(D_i\beta_j + D_j\beta_i\right). \tag{5}$$

The resulting components are

$$K_{xx} = -\frac{s}{2}\sqrt{\frac{r_0}{x^3}}, \tag{6}$$

$$K_{\theta\theta} = x\beta^x = s\sqrt{r_0 x}, \tag{7}$$

$$K_{\phi\phi} = x\beta^x \sin^2\theta, \tag{8}$$

$$K_{zz} = 0. \tag{9}$$

This remains subject to the final GRChombo K_{ij} sign convention.

4 Cartoon Contractions and Multiplicity

The angular contractions are equal:

$$\gamma^{\theta\theta} K_{\theta\theta} = \frac{1}{x^2}(x\beta^x) = \frac{\beta^x}{x}, \tag{10}$$

$$\gamma^{\phi\phi} K_{\phi\phi} = \frac{1}{x^2 \sin^2\theta}(x\beta^x \sin^2\theta) = \frac{\beta^x}{x}. \tag{11}$$

Thus the hidden cartoon component in this basic check is

$$K_{ww} = \frac{\beta^x}{x}. \tag{12}$$

For this project,

$$N_{\text{hidden}} = \texttt{GR\_SPACEDIM} - \texttt{CH\_SPACEDIM} = 4 - 2 = 2, \tag{13}$$

or in code notation,

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N_hidden = GR_SPACEDIM - CH_SPACEDIM = 4 - 2 = 2.
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The trace check is

$$K = K^x{}_x + K^z{}_z + 2K^w{}_w = \frac{3s}{2}\sqrt{\frac{r_0}{x^3}}. \tag{14}$$

5 Limits

The current scaffold verifies Christoffels, angular contractions, hidden multiplicity, and GP extrinsic-curvature formulas. It does not yet derive Ricci terms, CCZ4 RHS source terms, nontrivial h_{ww}, regularized small-x limits, or unit-test fixtures.